Lecture 7 - Interpreting, Estimating, and Applying the 1st Derivative and the 2nd Derivative

Garrett Credi

09/29/26

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- Start early! The later you start, the less time you'll have to ask me questions in office hours/via email!
- Review session - I'll hold an Exam 1 review session next Monday (from 6-8?) where you can come and ask any and all questions you have and I'll answer em!

Quiz Time!

Please get into neat rows, and spread out. Only then will I hand out the quiz. As per usual, no cheating (hopefully should go without saying).

Interpreting and Applying the First Derivative

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  and determine some information about $f(x)$ from $f'(x)$.

Increasing-ness, Decreasing-ness, and Stationary-ness I

We can use the derivative to determine whether a function is increasing/decreasing/stationary:

Increasing Function

Increasingness via the Derivative

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Increasingness via the Derivative

For a function $f(x)$ which is differentiable at $x = a$, increasingness at a is equivalent to $f'(a) > 0$. Similarly, decreasingness is equivalent to $f'(a) < 0$.

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When is a linear function increasing/decreasing? When the slope (which is $f'(a)$) is positive/negative!
In terms of instantaneous velocity: positive velocity means increasing position, resp. negative velocity means decreasing position.

Graph Sketching I

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And to sketch graphs of function $f(x)$ given graphs of their derivatives $f'(x)$:

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Notice how the increasingness/decreasingness of the *derivative* (e.g. the derivative of the derivative) informed us about the shape of the original function $f(x)$ - we'll see later in lecture this is due to the properties of the *second* derivative $f''(x)$.

Estimating The Derivative

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Second and Higher Derivatives

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The second derivative function is defined similarly.

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Similarly, just as the first derivative is equivalent to instantaneous *velocity* , the second derivative is equivalent to instantaneous *acceleration*!

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i.e. that is lies below it's tangent lines.

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Concavity in terms of the Second Derivative

A function $f(x)$ which is second differentiable at $x = a$

E.g.

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A function $f(x)$ which is second differentiable at $x = a$ is concave up at $x = a$ if and only if $f''(a) > 0$.

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Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!